STAT201 · STATISTICS · LECTURE 12

Simple Linear Regression

How to describe the relationship between two quantitative variables by a straight line, estimate it from data and assess its quality

Syllabus topic 12 · week 12 · 4 academic hours

LECTURE GOALS

- draw a scatter plot and write the simple linear regression model $y = \beta_0 + \beta_1 x + \varepsilon$;
- compute the OLS estimates b_0, b_1 by hand and through the correlation, interpret the slope and intercept;
- compute residuals, SSE, SSR, SST, R^2 and the residual standard error s_e;
- make predictions and understand the danger of extrapolation;
- analyse a residual plot and find violations of the model's conditions.

1 The model

Correlation (Lecture 6) measures the strength of a linear relationship but does not say *by how much* y changes on average when x grows by one unit. Regression does that. Suppose for n objects an *explanatory* variable x (predictor) and a *dependent* variable y (response) are measured.

THE SIMPLE LINEAR REGRESSION MODEL

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad \varepsilon_i \sim \text{independent, } \mathrm{E}[\varepsilon_i] = 0,\ \mathrm{Var}(\varepsilon_i) = \sigma^2. \tag{1.1}$$

β_1 is the change in the mean of y when x grows by one unit; β_0 is the mean of y at $x = 0$; ε is the random error (everything not explained by x). The parameters β_0, β_1 are unknown and are estimated from a sample.

We are modelling the *conditional mean* $\mathrm{E}[y \mid x] = \beta_0 + \beta_1 x$. This does not mean that x is the cause of y (Lecture 13).

2 The method of least squares

The line $\hat{y} = b_0 + b_1 x$ is chosen so that the sum of squared vertical deviations (the residuals $e_i = y_i - \hat{y}_i$) is minimal:

$$\min_{b_0, b_1} \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2. \tag{2.1}$$

The solution (setting the derivatives to zero):

$$b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}} = r\,\frac{s_y}{s_x}, \qquad b_0 = \bar{y} - b_1 \bar{x}. \tag{2.2}$$

From the second equality it follows that the line always passes through the point $(\bar{x}, \bar{y})$. From the formula $b_1 = r\, s_y / s_x$ it follows that the sign of the slope coincides with the sign of the correlation.

EXAMPLE 1. ADVERTISING AND SALES

Data from Lecture 6: advertising spend $x = 2, 3, 5, 6, 8, 10$ and sales $y = 15, 17, 22, 31, 33, 40$ (million UZS). $\bar{x} = 5.667$, $\bar{y} = 26.333$, $S_{xx} = 45.33$, $S_{yy} = 487.33$, $S_{xy} = 145.67$.

$$b_1 = \frac{145.67}{45.33} = 3.213, \qquad b_0 = 26.333 - 3.213 \cdot 5.667 = 8.125, \qquad \hat{y} = 8.125 + 3.213\,x.$$

Interpretation: each additional million of advertising is associated with an increase in sales of 3.21 million UZS on average. The intercept 8.1 is sales at zero spend (here $x = 0$ lies outside the data range 2–10, so it has no literal interpretation).

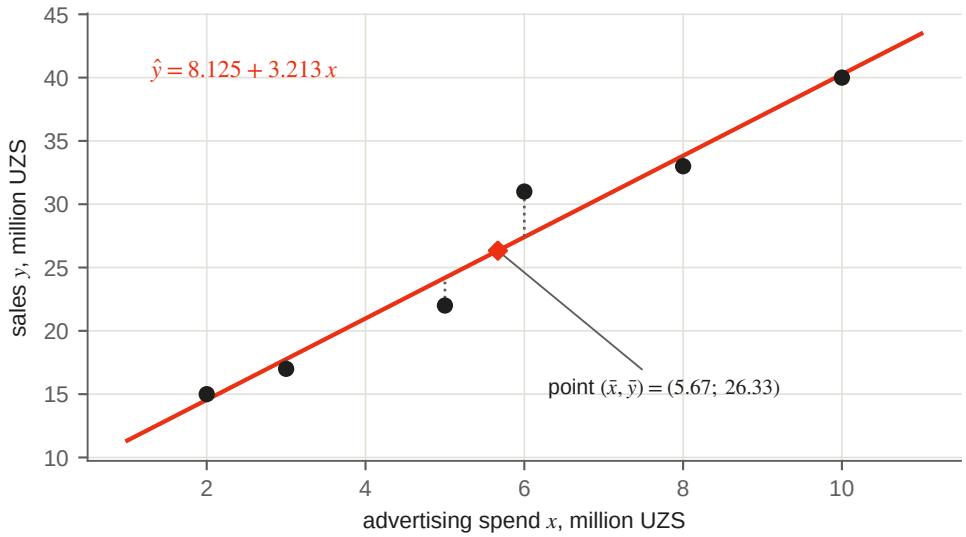


Fig. 1. The scatter plot and the OLS line. The dashed segments are the residuals; the line minimizes the sum of their squares. The diamond is the point $(\bar{x}, \bar{y})$ through which the line passes.

3 Residuals and goodness of fit

The predictions $\hat{y}_i = b_0 + b_1 x_i$: $14.55;\ 17.77;\ 24.19;\ 27.40;\ 33.83;\ 40.26$. The residuals $e_i = y_i - \hat{y}_i$: $0.45;\ -0.77;\ -2.19;\ 3.60;\ -0.83;\ -0.26$. Properties of OLS residuals: $\sum e_i = 0$ and $\sum x_i e_i = 0$ (check this!).

The total variability of y is decomposed into explained and unexplained:

$$\underbrace{\sum (y_i - \bar{y})^2}_{SST} = \underbrace{\sum (\hat{y}_i - \bar{y})^2}_{SSR} + \underbrace{\sum (y_i - \hat{y}_i)^2}_{SSE}, \qquad R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}. \tag{3.1}$$

COEFFICIENT OF DETERMINATION

R^2 is the share of the variation in y explained by the linear dependence on x. In simple regression $R^2 = r^2$.

In the example $SST = 487.33$, $SSE = 19.27$, $SSR = 468.06$; $R^2 = 0.9605$ ($= 0.980^2$): advertising explains 96% of the variation in sales. The residual standard error

$$s_e = \sqrt{\frac{SSE}{n-2}} = \sqrt{\frac{19.27}{4}} = 2.195 \tag{3.2}$$

shows the typical size of a prediction error in the units of y (we divide by $n-2$ because two parameters were estimated).

COMMON MISTAKE

A high R^2 proves neither causation nor the correctness of the model; a low R^2 does not mean there is no relationship (there may be a nonlinear relationship or a significant but small effect). Always look at the residual plot.

4 Prediction and extrapolation

The prediction of the mean at $x = 7$: $\hat{y} = 8.125 + 3.213 \cdot 7 = 30.62$ million UZS. This is acceptable, since 7 lies inside the range $[2; 10]$ (*interpolation*). The prediction at $x = 12$ gives 46.7, but this is *extrapolation*: beyond the data linearity is not guaranteed (market saturation, diminishing returns), and such a prediction cannot be trusted.

REMEMBER

The regression of y on x and the regression of x on y are different lines (except when $|r| = 1$). If you need to predict y, the explanatory variable is x; there is no symmetry in OLS. Units: b_1 is measured in units of y per unit of x; when the units of x are changed (million $\to$ thousand) b_1 changes, r and R^2 do not.

5 Diagnostics from the residuals

Conditions of the model: (1) linearity; (2) independence of errors; (3) constant variance of errors (homoscedasticity); (4) approximately normal errors (important for small n for inference, Lecture 13). The residual plot of e_i against $\hat{y}_i$ is examined: with a good fit the points are randomly scattered around zero with no structure.

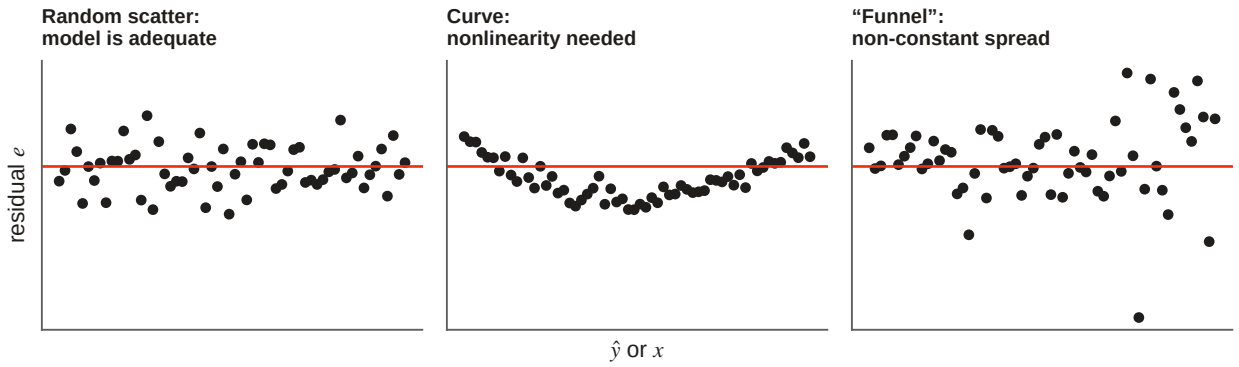


Fig. 2. Three typical residual patterns. A curve means that the relationship is nonlinear, a “funnel” that the spread grows with x (heteroscedasticity).

5.1 Outliers and influential observations

A point with a large residual is an outlier in y. A point with an unusual value of x (far from $\bar{x}$) has high leverage and may “pull” the line. Always find out the reason (a typing error, a special case) and compare results with and without the point (as in Lecture 1 for branch 12), rather than deleting data for the sake of a nice line.

6 Key terms

Term	*In Russian*	Meaning
Response (dependent) variable, y	*Зависимая переменная*	the variable being explained
Predictor (explanatory) variable, x	*Объясняющая переменная*	the factor
Ordinary least squares (OLS)	*МНК*	minimum of $\sum e_i^2$
Residual	*Остаток*	$e_i = y_i - \hat{y}_i$
Coefficient of determination, R^2	*Коэффициент детерминации*	$SSR/SST = r^2$
Residual standard error	*Стандартная ошибка регрессии*	$s_e = \sqrt{SSE/(n-2)}$
Extrapolation	*Экстраполяция*	a prediction outside the range of x
Homoscedasticity	*Гомоскедастичность*	constant variance of errors

Formulas of Lecture 12

$$b_1 = \frac{S_{xy}}{S_{xx}} = r\frac{s_y}{s_x}, \qquad b_0 = \bar{y} - b_1\bar{x}, \qquad e_i = y_i - \hat{y}_i;$$

$$SST = SSR + SSE, \qquad R^2 = \frac{SSR}{SST} = r^2, \qquad s_e = \sqrt{\frac{SSE}{n-2}}.$$

7 Practice problems

> Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix “Statistical Tables”.

Problem 1 [A] The regression equation of delivery cost y (thousand UZS) on distance x (km): $\hat{y} = 10 + 2.5x$.

(a) Interpret b_1 and b_0.
(b) Find the prediction at $x = 8$ km and the residual for the observation $(8;\ 32)$.

Problem 2 [A] Given: $\bar{x} = 40$, $\bar{y} = 100$, $s_x = 10$, $s_y = 30$, $r = 0.6$. Find the regression equation of y on x and R^2.

Problem 3 [A] Study time x (h): $1, 2, 3, 4, 5$; scores y: $50, 55, 65, 70, 85$. Find the regression equation, the prediction at $x = 4$ and the residual for this observation.

Problem 4 [B] For the data of the previous problem find all predictions and residuals, check $\sum e_i = 0$, find SSE, SST, R^2 and s_e.

Problem 5 [B] Apartment area x (m^2): $40, 50, 60, 80, 100$; price y (thousand USD): $38, 45, 50, 68, 84$.

(a) Find the regression equation.
(b) Find R^2 and s_e.
(c) Predict the price of a 70 m^2 apartment. Can the model be used for a 200 m^2 apartment?

Problem 6 [B] For a regression on 22 observations $SST = 500$, $SSE = 120$.

(a) Find SSR and R^2.
(b) Find $|r|$ and s_e.
(c) What can be said about the sign of r?

Problem 7 [B] In the regression of Lecture 12 ($\hat{y} = 8.125 + 3.213x$, x and y in million UZS) express x in thousands of UZS ($x' = 1000x$). How do b_0, b_1, r, R^2 and s_e change?

Problem 8 [B] Using the regression of Lecture 12 an analyst predicts sales for advertising of 25 million UZS and gets $\hat{y} = 88.5$. What is wrong with this prediction? What should he do?

Problem 9 [B] For the data of Lecture 12 the regression of y on x gives $b_1 = 3.213$. (a) Find the slope of the regression of x on y. (b) Check that the product of the two slopes equals r^2. Can x be found from a predicted y simply by solving the equation $y = b_0 + b_1 x$ for x?

Problem 10 [C] Derive the OLS estimates: find b_0 and b_1 minimizing $Q = \sum (y_i - b_0 - b_1 x_i)^2$.

Problem 11 [C] Show that (a) $\sum e_i = 0$; (b) $R^2 = r^2$ in simple regression.